Clustering and Differential Analysis

Yongjin Park

05 March 2026

Roadmap: From biology to methodology

14: Biological Networks

- Protein-protein, genetic interactions
- Cell-cell interaction networks
- Network as a tool: propagation

15: Clustering

- Graph-based clustering
- Single-cell example
- When clustering goes wrong

Today's lecture

1. Graph-based clustering
2. Single-cell clustering in practice
3. When clustering goes wrong
4. Discussion: How should we test cell-cell communication?
5. Appendix

Last time: networks encode relationships

- We learned how to build and visualize networks
- Protein-protein interactions, genetic interactions, cell-cell communication
- Today: how do we find **groups** within a network?

Recap: `igraph` in R

```r
library(igraph)
```

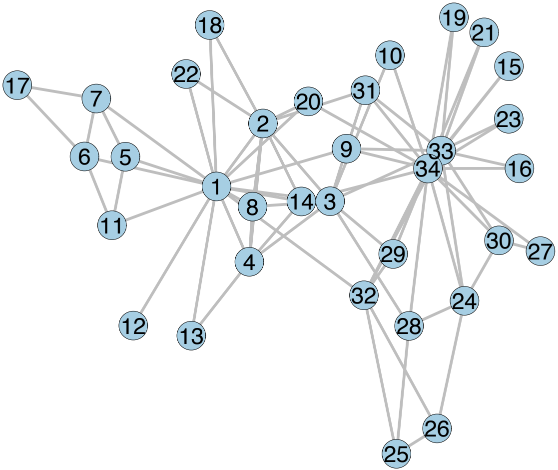
```r
karate <- .read.online("http://www-personal.umich.edu/~mejn/netdata/karate.zip")
karate
```

```
## IGRAPH cd4b8ea U--- 34 78 --
## + attr: id (v/n)
## + edges from cd4b8ea:
##  [1]  1-- 2  1-- 3  2-- 3  1-- 4  2-- 4  3-- 4  1-- 5  1-- 6  1-- 7  5-- 7
## [11]  6-- 7  1-- 8  2-- 8  3-- 8  4-- 8  1-- 9  3-- 9  3--10  1--11  5--11
## [21]  6--11  1--12  1--13  4--13  1--14  2--14  3--14  4--14  6--17  7--17
## [31]  1--18  2--18  1--20  2--20  1--22  2--22 24--26 25--26  3--28 24--28
## [41] 25--28  3--29 24--30 27--30  2--31  9--31  1--32 25--32 26--32 29--32
## [51]  3--33  9--33 15--33 16--33 19--33 21--33 23--33 24--33 30--33 31--33
## [61] 32--33  9--34 10--34 14--34 15--34 16--34 19--34 20--34 21--34 23--34
## [71] 24--34 27--34 28--34 29--34 30--34 31--34 32--34 33--34
```

- $n = |V|$: number of vertices, $m = |E|$: number of edges
- d_v: degree of vertex v (how many neighbours?)

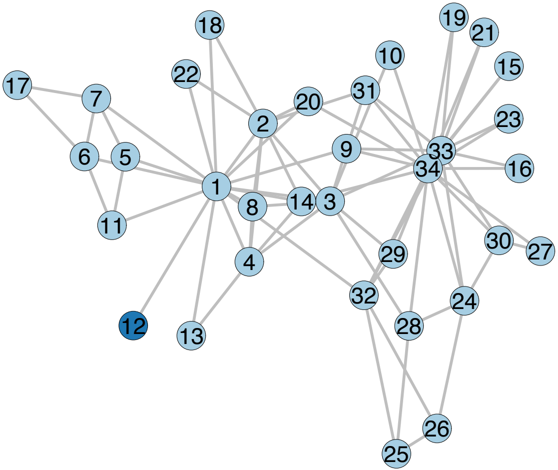
A classic example: Karate club (Zachary 1977)

- 34 members, edges = friendships
- The club split into two factions after a dispute
- Can we recover this split from the network alone?
- Who are the “influencers”?
  - **1**: the instructor (“Mr. Hi”)
  - **34**: the president (John A.)



Idea: split the network by cutting edges

- Consider two groups — find a set of edges to cut so the graph splits into two disconnected components
- **Min-cut**: the fewest edges to remove



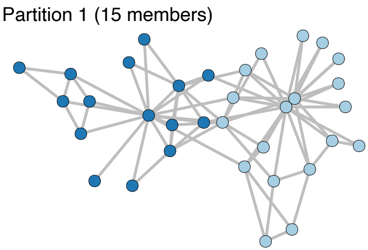
A trivial solution?

Split G by the fewest edges between two parties

```
.cuts <-
    as.directed(karate) %>%
    st_min_cuts(source = "1", target = c("33","34"))
.cuts$value / 2  # number of edges to cut
```

```
## [1] 5
```

Min-cut partitions from source to target



- Multiple solutions with the same min-cut $= 5$ edges
- But min-cut requires knowing **source and target** — can we find groups without labels?

The network as an adjacency matrix A

- Row/column: a member
- Non-zero element: friendship
- Can you see group structures?

$$A_{ij} = \begin{cases} 1 & i,j \text{ connected} \\ 0 & \text{otherwise} \end{cases}$$

Stochastic block model: a generative model of A?

- An adjacency matrix A can be **data**

$$A_{ij} \sim \text{Bernoulli}\left(\sum_{k,k'} \theta_{kk'} Z_{ik} Z_{jk}\right)$$

where

$$Z_{ik} = \begin{cases} 1 & i \in \text{ cluster/block/membership } k \\ 0 & \text{otherwise} \end{cases}$$

and

$$\theta_{kk'} = P(A = 1 | \text{between communities } k \text{ and } k')$$

- **Goal**: factorize $A \approx Z^{\top} \Theta Z$

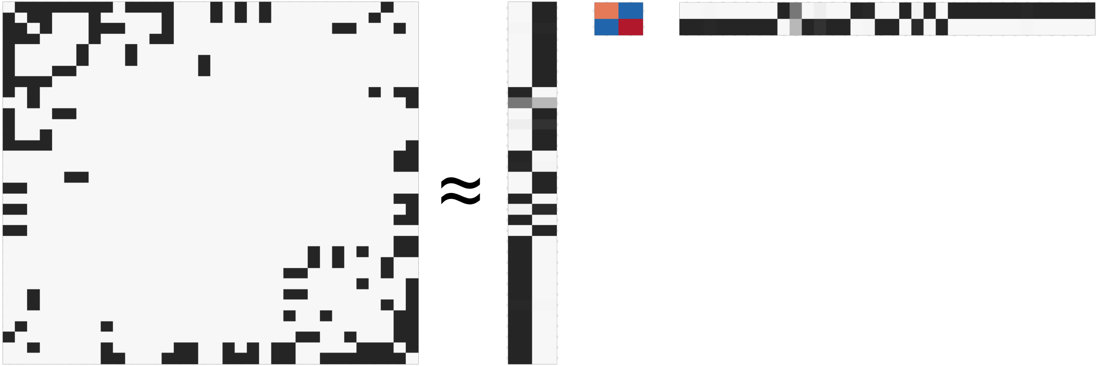
Fitting the stochastic block model

```r
library(hsblock)
.out <- fit.hsblock(A, "bernoulli", vbiter=100,
                    inner.iter=10, verbose=FALSE)
Z <- as.matrix(.out$Z)
```

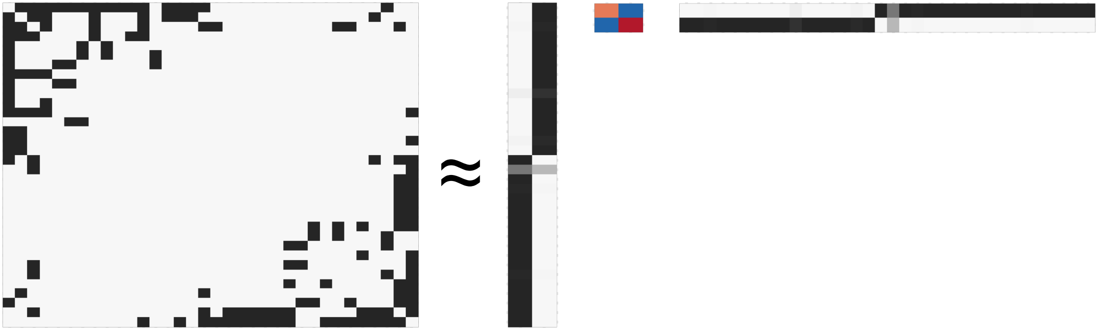
```r
n <- ncol(Z)
Edges <- 0.5 * Z %*% A %*% t(Z)
.ones <- matrix(1,n,n)
diag(.ones) <- 0
Tot <- Z %*% .ones %*% t(Z)/2
Theta <- Edges / Tot
```

- $\mathbf{Z}$: 2 clusters $\times$ 34 vertices
- Θ: edge probabilities

Matrix factorization view: $A \approx Z^\top \Theta Z$

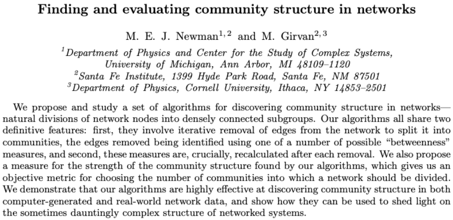


Reordering by cluster reveals the block structure



- Sorting vertices by cluster membership makes the block structure visible
- Each block shows edge density within/between groups

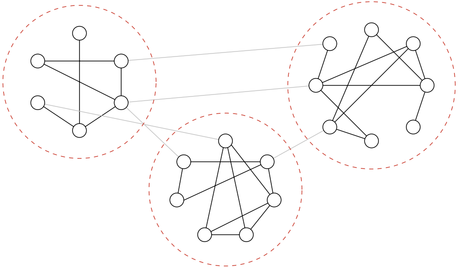
How do we know clustering is good enough?


Finding and evaluating community structure in networks

M. E. J. Newman[1,2] and M. Girvan[2,3]

[1] *Department of Physics and Center for the Study of Complex Systems, University of Michigan, Ann Arbor, MI 48109–1120*
[2] *Santa Fe Institute, 1399 Hyde Park Road, Santa Fe, NM 87501*
[3] *Department of Physics, Cornell University, Ithaca, NY 14853–2501*

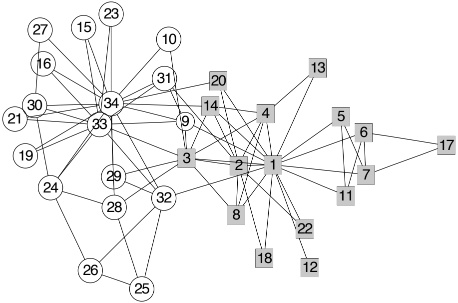
We propose and study a set of algorithms for discovering community structure in networks—natural divisions of network nodes into densely connected subgroups. Our algorithms all share two definitive features: first, they involve iterative removal of edges from the network to split it into communities, the edges removed being identified using one of a number of possible “betweenness” measures, and second, these measures are, crucially, recalculated after each removal. We also propose a measure for the strength of the community structure found by our algorithms, which gives us an objective metric for choosing the number of communities into which a network should be divided. We demonstrate that our algorithms are highly effective at discovering community structure in both computer-generated and real-world network data, and show how they can be used to shed light on the sometimes dauntingly complex structure of networked systems.



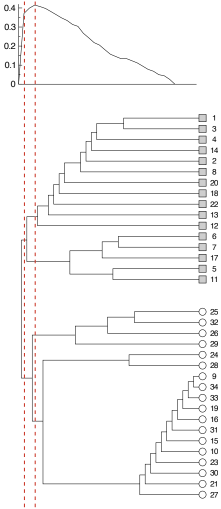
It’s not always as perfect as this. . .

- cluster ≈ densely connected subgraph
- but how dense?

In a dendrogram of clusters, where should we cut?



This is perhaps what happened after the split...



- **What will be the proper score?**
- The underlying combinatorial optimization is one of the hardest CS problems.

What is modularity? (Newman and Girvan 2004)

Given a partition of vertices into groups, **modularity** Q measures how many more edges fall *within* groups than expected by chance:

$$Q = \frac{1}{2m} \sum_{ij} \left[\mathbf{A}_{ij} - \frac{d_i d_j}{2m} \right] \delta(c_i, c_j)$$

- A_{ij}: adjacency matrix, d_i: degree of vertex i, m: total edges
- $\delta(c_i, c_j) = 1$ if vertices i, j are in the same group
- $Q \in [-0.5, 1]$; higher $=$ stronger community structure
- The null model: edges are distributed proportionally to degree

Why $d_i d_j / 2m$?

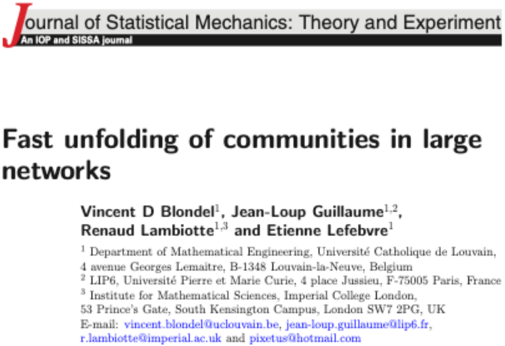
- Each vertex i has d_i "stubs" (half-edges). Total stubs: $\sum_i d_i = 2m$.
- Pick a stub from i. It connects to one of $2m - 1 \approx 2m$ other stubs.
- Vertex j has d_j stubs, so:

$$Pr(\text{stub from } i \text{ lands on } j) = \frac{d_j}{2m}$$

- Vertex i has d_i stubs, so the expected number of edges between i and j:

$$\mathbb{E}[A_{ij}] = \frac{d_i\, d_j}{2m}$$

Louvain: Greedy optimization of the modularity score



Journal of Statistical Mechanics: Theory and Experiment

An IOP and SISSA journal

Fast unfolding of communities in large networks

Vincent D Blondel[1], Jean-Loup Guillaume[1,2], Renaud Lambiotte[1,3] and Etienne Lefebvre[1]

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[2] LIP6, Université Pierre et Marie Curie, 4 place Jussieu, F-75005 Paris, France

[3] Institute for Mathematical Sciences, Imperial College London, 53 Prince's Gate, South Kensington Campus, London SW7 2PG, UK

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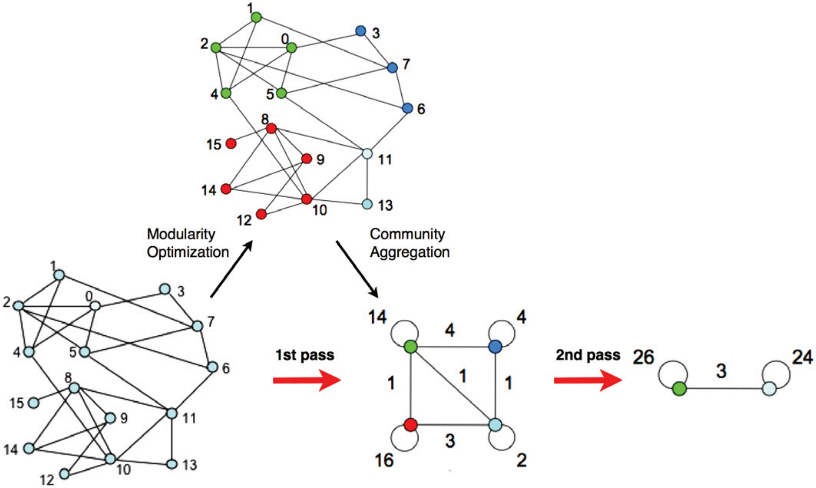
(Blondel et al. 2008):

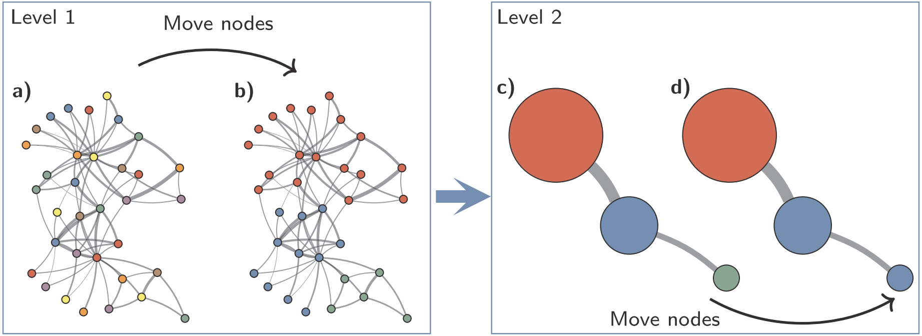
1. Each vertex starts as its own community
2. Move vertices to neighbour's community if it increases Q
3. Aggregate communities into super-nodes
4. Repeat until Q converges

Louvain algorithm: vertices $\rightarrow$ “super” nodes



(Blondel et al. 2008)

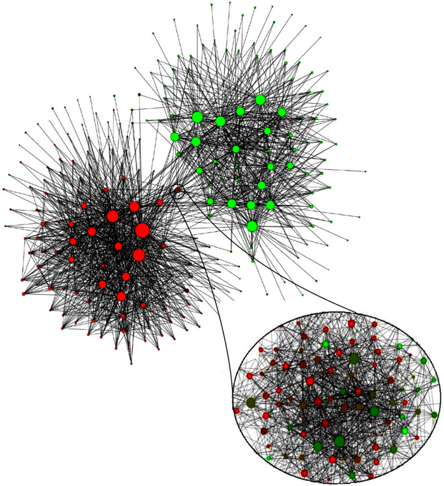
Louvain deals with the “colours” of “super” nodes



Key idea: Keep coarsening a large graph into a smaller one

(Traag et al. 2019)

Louvain can easily scale up to a million vertices



(Blondel et al. 2008)

Louvain's supernode with disconnected components


SCIENTIFIC REPORTS

OPEN

From Louvain to Leiden: guaranteeing well-connected communities

Received: 20 November 2018
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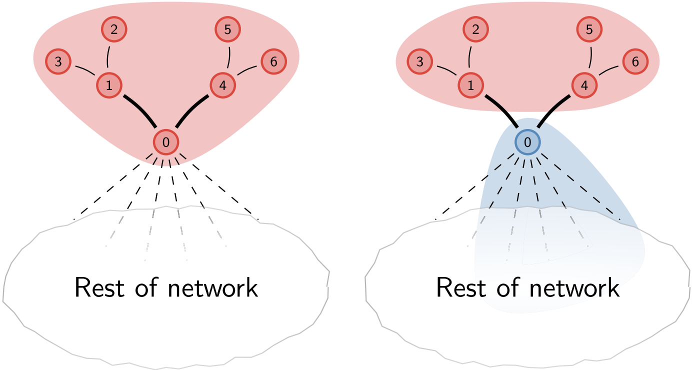
Published online: 26 March 2019

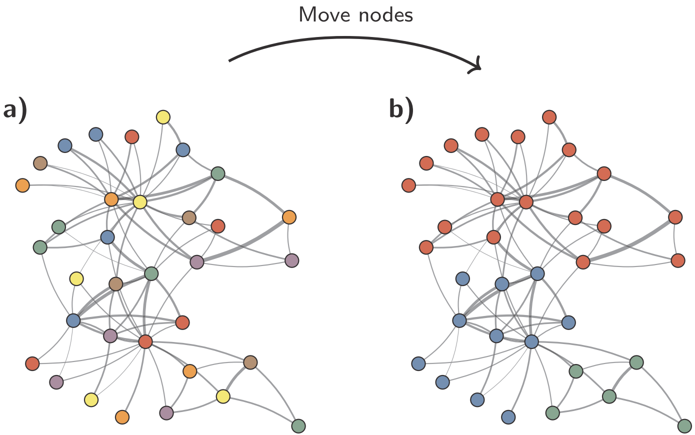
V. A. Traag, L. Waltman & N. J. van Eck

Community detection is often used to understand the structure of large and complex networks. One of



(Traag et al. 2019)

Leiden “refines” to ensure connected within a cluster



(Traag et al. 2019)

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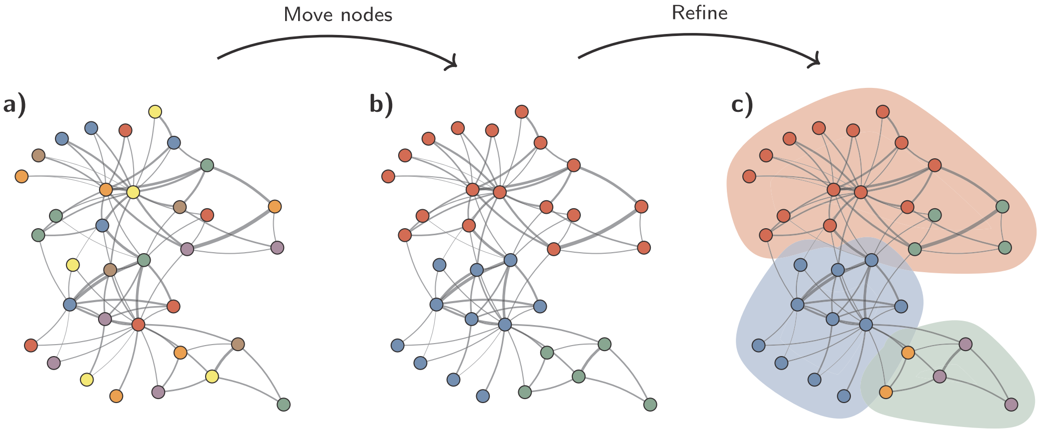
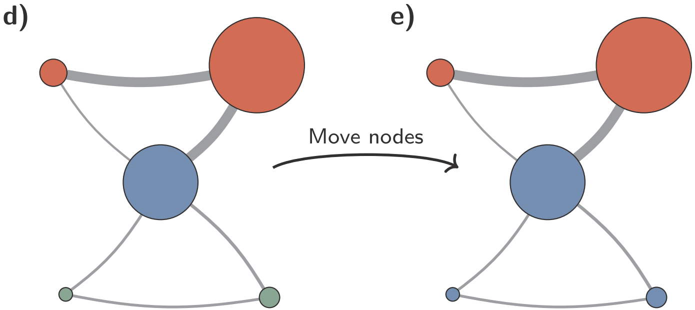


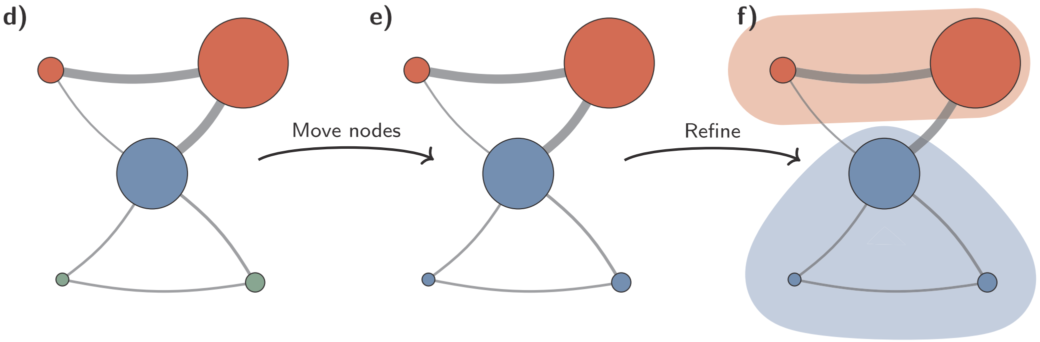
(Traag et al. 2019)

Leiden "refines" to ensure connected within a cluster



(Traag et al. 2019)

Leiden "refines" to ensure connected within a cluster

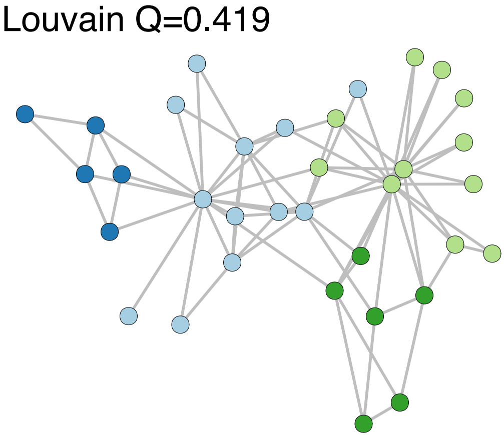


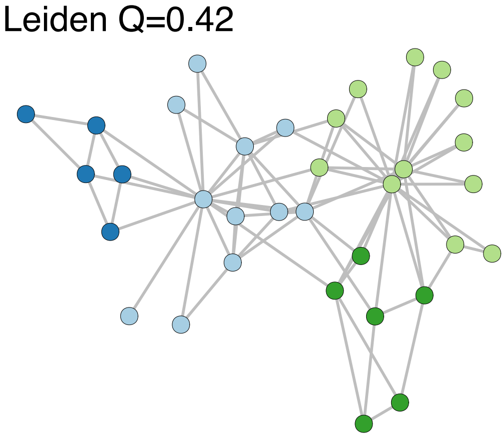
(Traag et al. 2019)

Leiden vs. Louvain

```r
cl.louvain <- cluster_louvain(karate)
cl.leiden  <- cluster_leiden(karate,
    objective_function = "modularity")
```

Leiden vs. Louvain



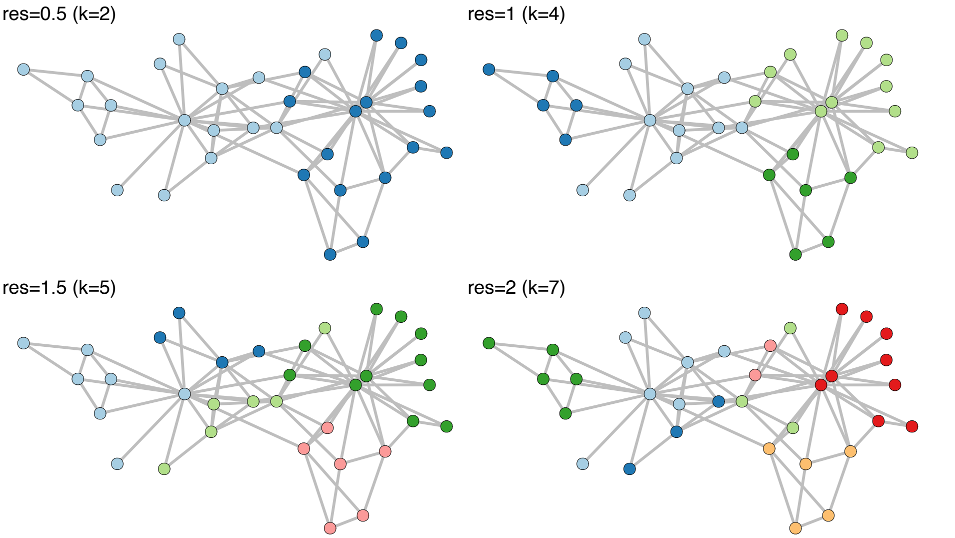


What should we learn from Louvain/Leiden?

When n is large. . .

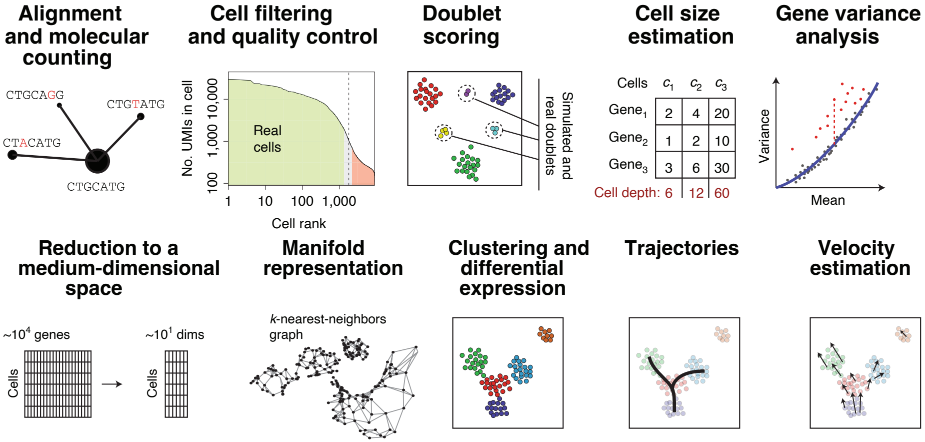
- What shall we do?
- Any counter examples for Leiden?

Side note: The resolution parameter matters



Today's lecture

Recall: the standard single-cell workflow



(Kharchenko 2021)

Clustering the PDAC data

```r
pdac <- Read10X(data.dir = data.dir)
pdac <- CreateSeuratObject(counts = pdac,
    project = "PDAC_TISSUE_1",
    min.cells = 3, min.features = 200)

pdac <- pdac |>
    NormalizeData() |>
    FindVariableFeatures(nfeatures = 2000) |>
    ScaleData() |>
    RunPCA(npcs = 30) |>
    FindNeighbors(dims = 1:20) |>
    RunUMAP(dims = 1:20)
```

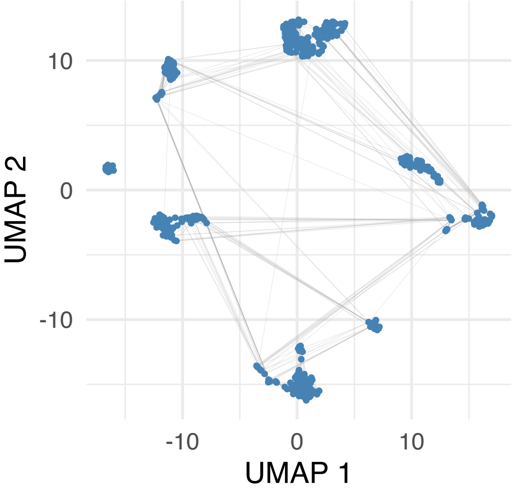
- Same PDAC Tissue 1 dataset from Lecture 8 (Steele et al. 2020)
- `FindNeighbors()` builds the **KNN** and **SNN** graphs in PCA space

Building the KNN graph

```r
knn.mat <- pdac@graphs$RNA_nn
dim(knn.mat)
```

```
## [1] 995 995
```

- `FindNeighbors(dims = 1:20)` connects each cell to its $k = 20$ nearest neighbours in PCA space
- $A_{ij} = 1$ if cell j is among k-NN of cell i

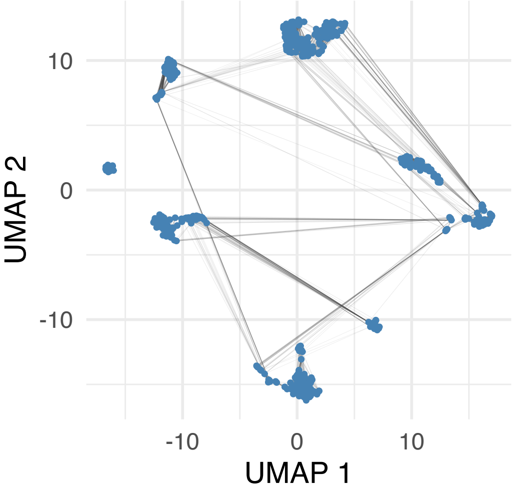


Shared neighbours graph

- **SNN**: weight edges by shared neighbour overlap (Jaccard index)

$$w_{ij} = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$$

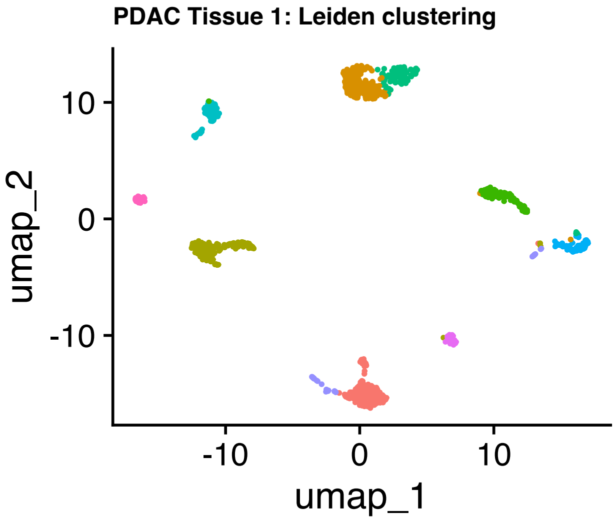
- Cells sharing many neighbours get stronger connections
- This is what Leiden/Louvain clusters on



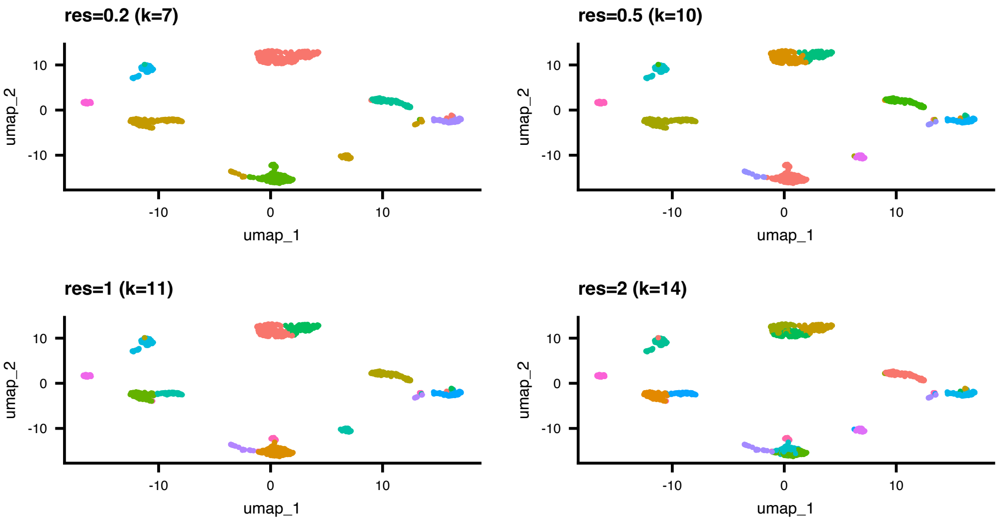
Clustering the PDAC data

```r
pdac <- FindClusters(
    pdac,
    algorithm = 4,
    resolution = 0.5)
```

- Leiden on the SNN graph
- 10 clusters found
- Resolution = 0.5



Effect of the resolution parameter



The resolution parameter in the modularity score

The **resolution parameter** γ scales the null model:

$$Q_\gamma = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \gamma \frac{d_i d_j}{2m} \right] \delta(c_i, c_j)$$

- $\gamma > 1$: penalizes large clusters more $\rightarrow$ **more, smaller** clusters
- $\gamma < 1$: tolerates sparser communities $\rightarrow$ **fewer, larger** clusters
- $\gamma = 1$: standard modularity

Today's lecture

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2. Single-cell clustering in practice
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Are clusters real or artifacts of the algorithm?

nature methods

Article https://doi.org/10.1038/s41592-023-01933-9

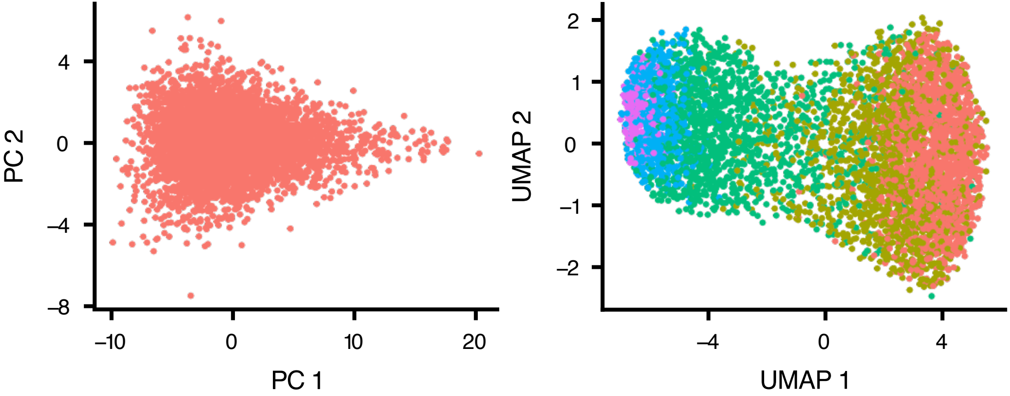
Significance analysis for clustering with single-cell RNA-sequencing data

Received: 1 August 2022

Isabella N. Grabski [1], Kelly Street [2] & Rafael A. Irizarry [3]

(Grabski et al. 2023)

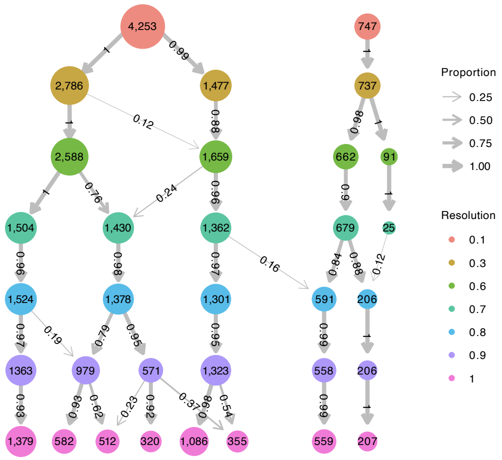
Clustering a single population finds spurious clusters



Simulated data with **no true subpopulations**. PCA (left) shows one blob; Leiden (right) still finds clusters.

(Grabski et al. 2023)

Higher resolution keeps splitting, but are they real?



Each node = a cluster at a given resolution. Arrows show how cells redistribute as resolution increases.

Thin arrows = small proportion of cells moving → potentially spurious splits.

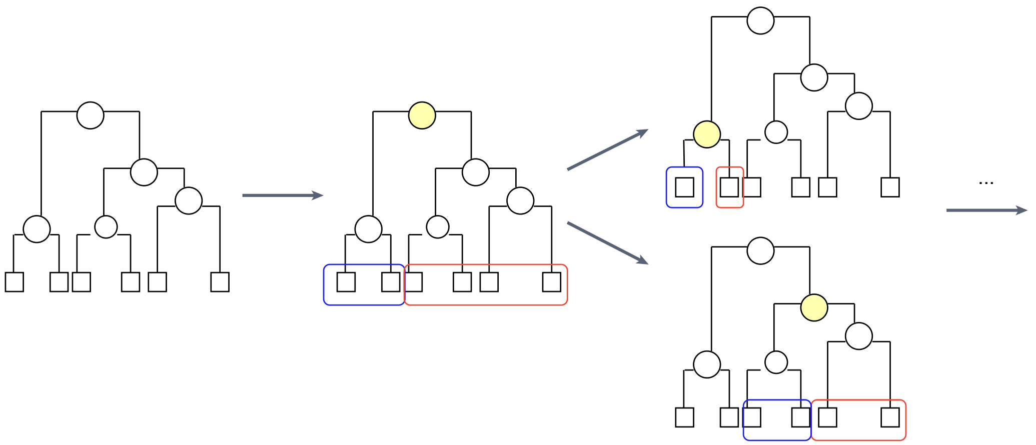
(Grabski et al. 2023)

sc-SHC: recursively test each split for significance

Hierarchically cluster cells (or pre-computed clusters)

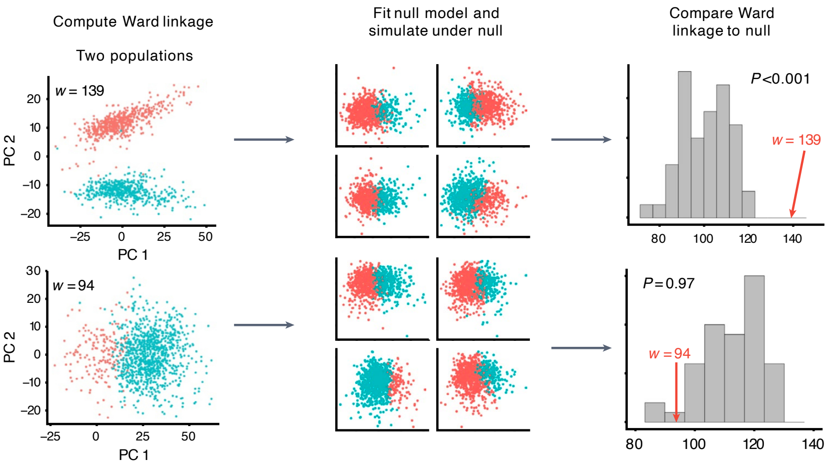
Test split at the root

Stop if fail to reject; otherwise proceed recursively



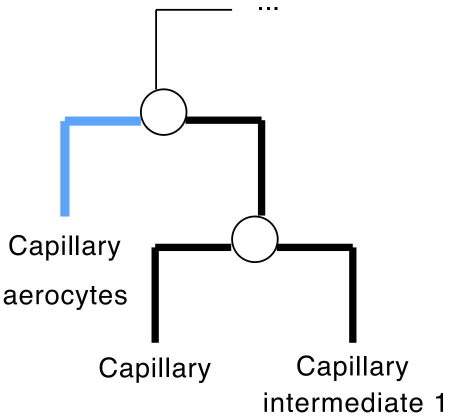
(Grabski et al. 2023)

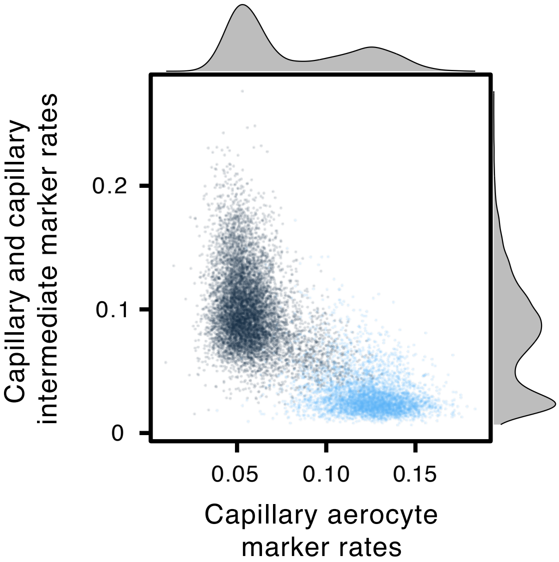
Can we compute the p-value of each split?



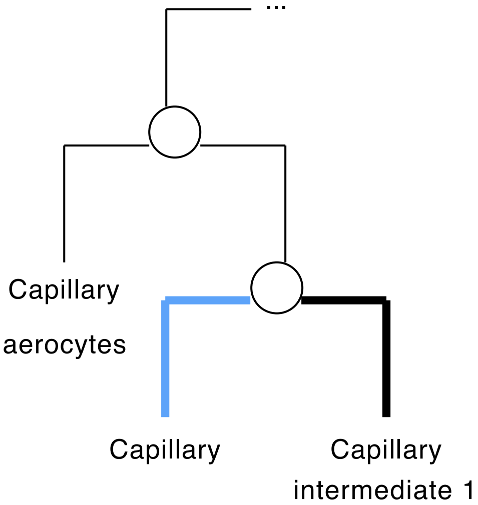
Top: two real populations → observed w far from null ($P < 0.001$). Bottom: one population → w typical under null ($P = 0.97$). (Grabski et al. 2023)

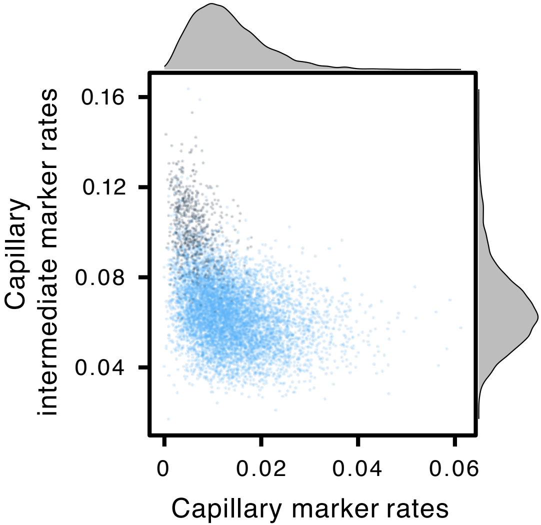
Capillary aerocytes are a distinct subtype





Not all splits are real: capillary subtypes overlap





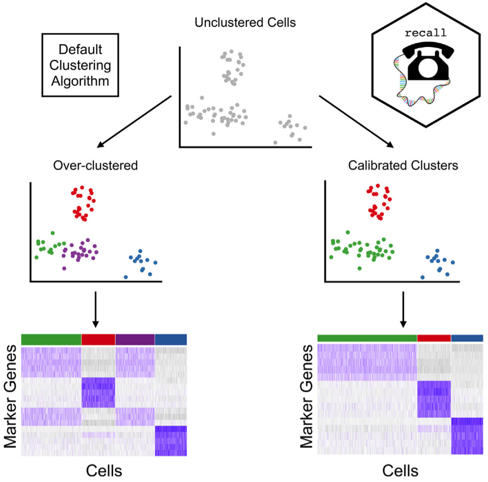
RECALL combats against an over-clustering problem

Problem: Leiden with a high resolution parameter → too many clusters → spurious DE genes

RECALL (calibrated clustering with artificial variables):

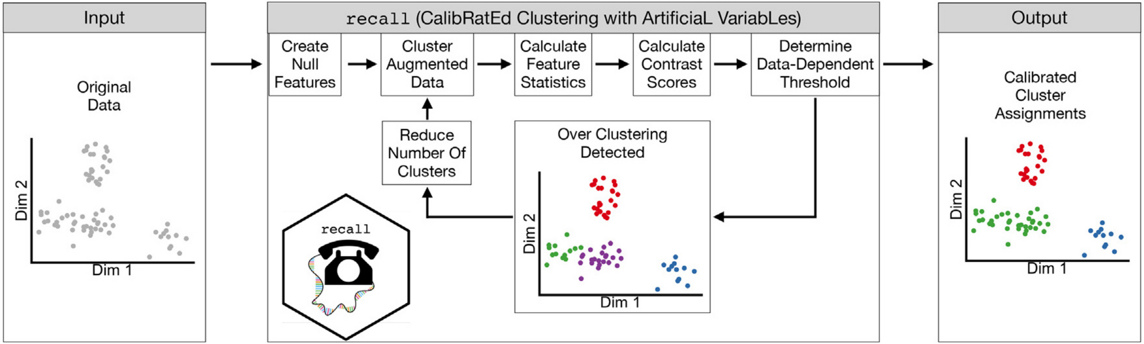
1. **Augment** the expression matrix with **artificial null genes** — synthetic variables that match the data distribution but have no association with cell types
2. **Cluster** cells using both real and artificial genes
3. **Test for DE** between clusters, for both real and artificial genes
4. Since artificial genes are known nulls, their test statistics reveal the **double-dipping effect**
5. If real genes are not more significant than artificial ones → **merge** those clusters
6. Repeat until clusters are well-calibrated

Default clustering over-clusters

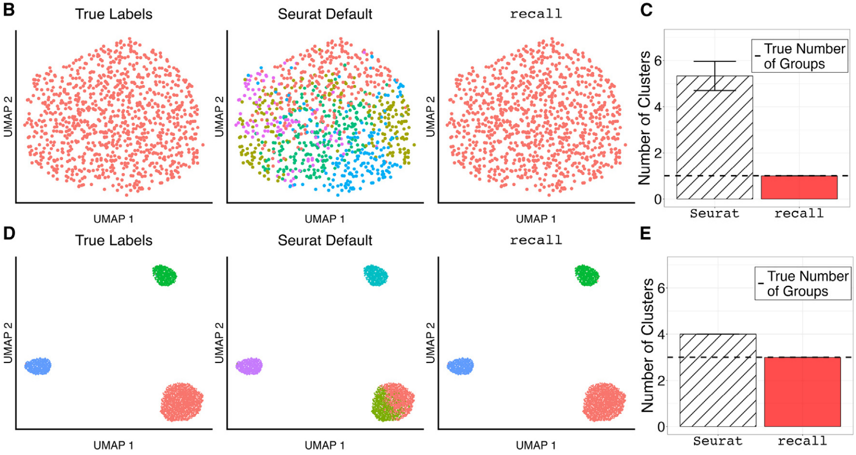


(DenAdel et al. 2025)

RECALL: iteratively reduce resolution until calibrated

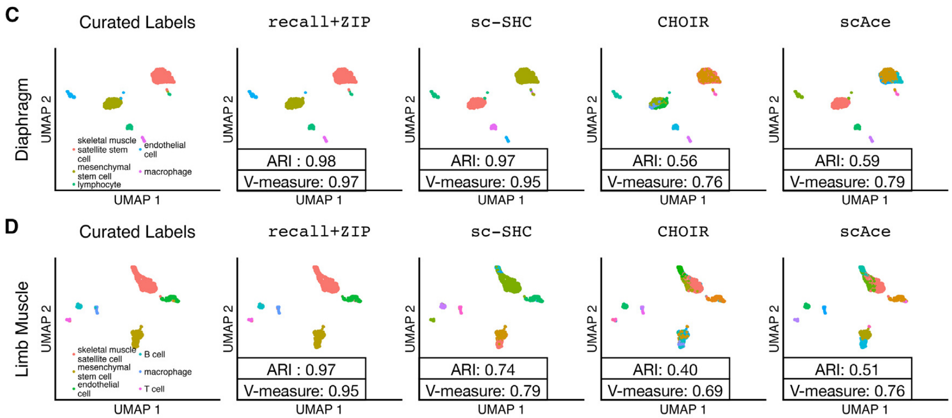


RECALL correctly identifies true clusters



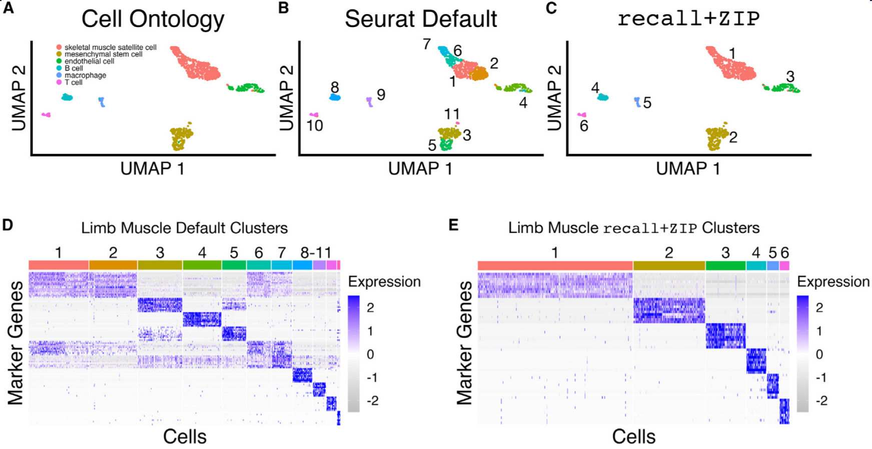
B–C: one population — Seurat finds about 5, RECALL finds 1. D–E: three populations — both recover 3. (DenAdel et al. 2025)

RECALL outperforms alternatives on real tissue data



Diaphragm and limb muscle datasets. ARI and V-measure vs curated labels. (DenAdel et al. 2025)

Seurat default over-splits; RECALL matches cell ontology



Limb muscle: Seurat finds 11 clusters, RECALL finds 6 matching the 6 annotated cell types. Heatmaps show cleaner marker gene blocks. (DenAdel et al. 2025)

How does RECALL generate artificial null genes?

For each gene j, fit a **zero-inflated Poisson** (ZIP) to the observed counts:

$$P(X_j = 0) = \pi_{0j} + (1 - \pi_{0j})\, e^{-\lambda_j}, \quad P(X_j = x) = (1 - \pi_{0j})\, \frac{\lambda_j^x\, e^{-\lambda_j}}{x!}$$

Estimate parameters from the data:

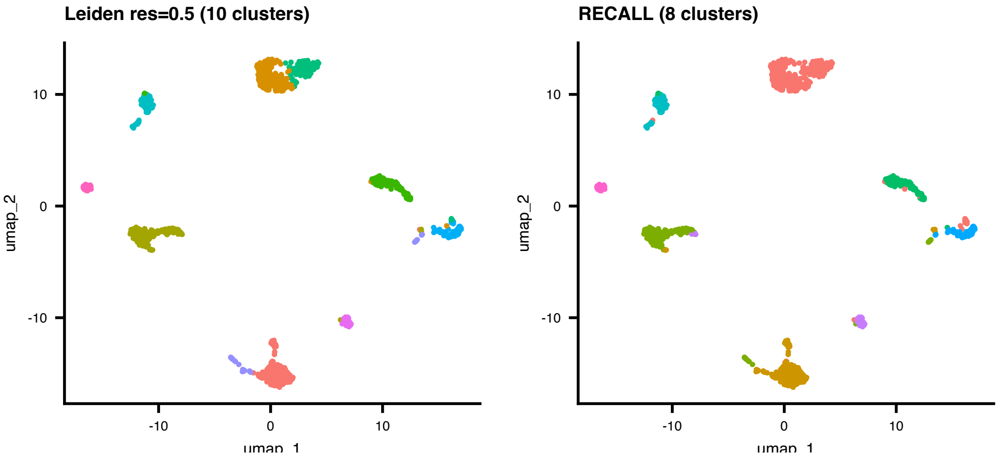
- $\hat{\lambda}_j$ from the mean $\bar{x}_j$ and zero-fraction r_{0j} via the Lambert W function
- $\hat{\pi}_{0j} = 1 - \bar{x}_j / \hat{\lambda}_j$

Then sample synthetic null genes: $\tilde{x}_j \sim \mathrm{ZIP}(\hat{\pi}_{0j},\, \hat{\lambda}_j)$

Each null gene matches the **marginal** distribution (mean, zero-inflation) of the real gene.

RECALL on the PDAC data

```r
library(recall)
pdac.recall <- FindClustersRecall(pdac,
    algorithm = "leiden",
    resolution_start = 0.5, dims = 1:20)
```

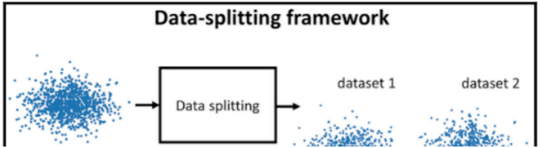


Cell Systems

Report

Valid Post-clustering Differential Analysis for Single-Cell RNA-Seq

Graphical Abstract



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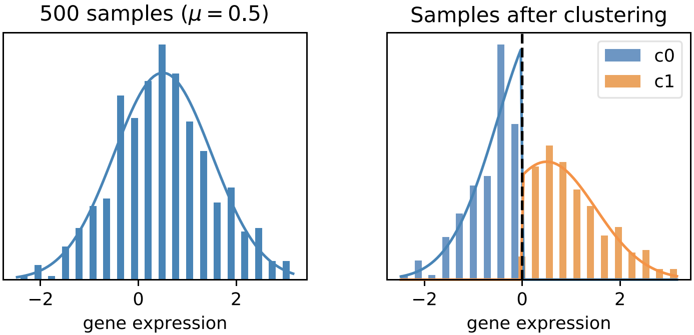
- Cluster cells using gene expression $\rightarrow$ find marker genes that differ between clusters
- But clusters were *defined* to maximize gene expression differences
- This is **selection bias**: the same data used to form and evaluate the hypothesis

(Zhang et al. 2019)

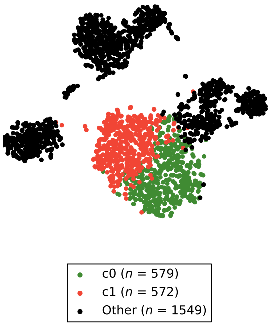
Clustering creates artificial DE even from one population



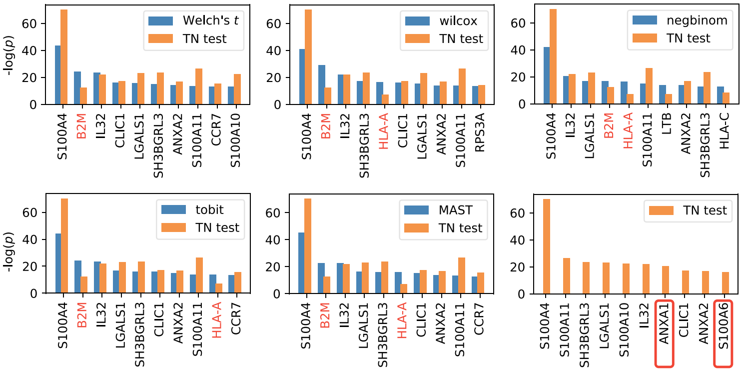
500 cells from **one** distribution ($\mu = 0.5$). After k-means, the two clusters look different — but the difference is an artefact. (Zhang et al. 2019)

Standard DE tests are inflated; corrected tests are not

10x PBMC dataset



Most significant genes for Seurat differential expression tests (c0 v c1)



Blue = standard test, orange = truncated normal (TN) corrected test. Red gene names = housekeeping genes that should not be DE. (Zhang et al. 2019)

Clustering then testing $\rightarrow$ inflated significance

Even when there is **no true grouping**, standard DE after clustering produces many "significant" genes:

- A "double dipping" problem
- The clustering algorithm forces cells into groups
- Cells assigned to the same cluster are more similar *by construction*
- Standard tests (e.g., Wilcoxon, t-test) ignore this selection step $\rightarrow$ **inflated p-values**

The double-dipping problem

We observe a count matrix X (cells $\times$ genes), where $X_{ij} \sim \text{Poisson}(\lambda_{ij})$

Standard pipeline (double dipping):

- Use X to estimate clusters $\hat{c}_1, \ldots, \hat{c}_n$
- Use X again to test: "Is gene j differentially expressed between clusters?"
- The test does not account for the fact that clusters were chosen to make genes look different

Result: under the null (λ_{ij} does not depend on cell type), we still get many small p-values

(Neufeld et al. 2023)

Inference after latent variable estimation for single-cell RNA sequencing data

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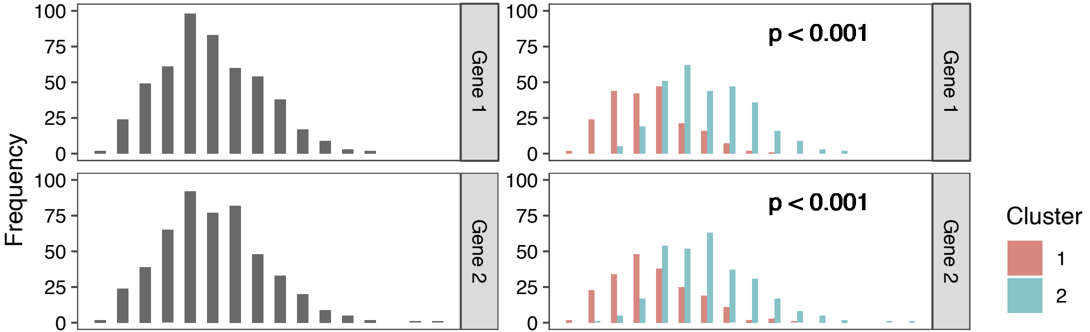
Null genes look DE after clustering: both $p < 0.001$



Two genes drawn from the **same** Poisson distribution across all cells. After clustering (right), both appear significantly DE ($p < 0.001$). (Neufeld et al. 2023)

Count splitting (Neufeld et al. 2023)

Key insight: Poisson thinning creates independent splits

If $X_{ij} \sim \text{Poisson}(\lambda_{ij})$, then for any $\epsilon \in (0, 1)$:

$$X_{ij}^{\text{train}} \mid X_{ij} \sim \text{Binomial}(X_{ij}, \epsilon), \quad X_{ij}^{\text{test}} = X_{ij} - X_{ij}^{\text{train}}$$

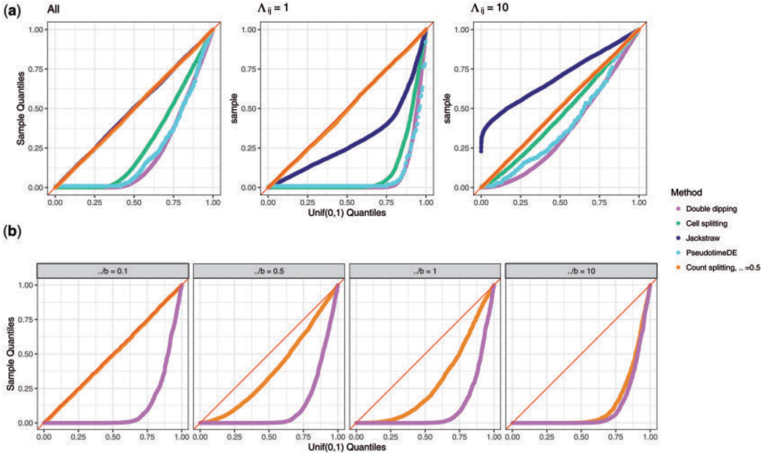
Then $X^{\text{train}} \perp X^{\text{test}}$ (they are **independent**!)

The recipe:

1. Split X into X^{train} and X^{test} via Poisson thinning
2. Cluster using X^{train} only
3. Test for DE using X^{test} only

This controls Type I error because the test data is independent of the clustering.

Count splitting controls Type I error

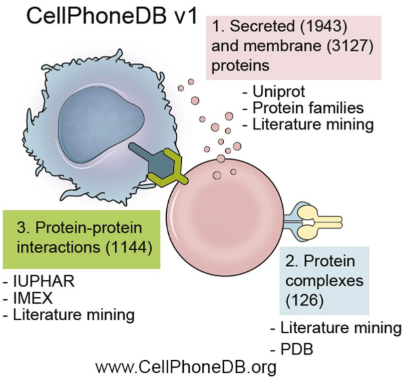


- QQ plots under null: double dipping (purple) inflates p-values
- Count splitting (orange) stays on the diagonal.
- Count splitting across signal-to-noise ratios.

(Neufeld et al. 2023)

Today's lecture

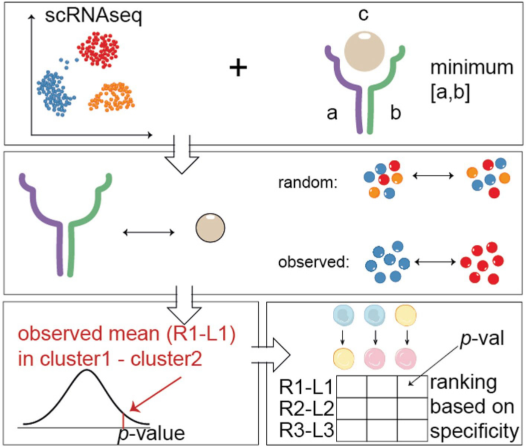
CellPhoneDB infers communication from clusters



1. **Cluster** cells into discrete types (e.g., Leiden)
2. **Look up** known ligand-receptor (L-R) pairs from a curated database
3. For each L-R pair (l, r) and each cell-type pair (A, B):
   - Compute the **mean expression** of l in A and r in B
   - Score $= \text{mean}(\bar{x}_l^A, \bar{x}_r^B)$
4. **Permutation test**: randomly shuffle cell-type labels $\rightarrow$ null distribution of scores
5. Report pairs with $p < 0.05$

(Vento-Tormo et al. 2018; Armingol et al. 2021; Efremova et al. 2020)

CellPhoneDB output: significant L-R pairs



(Vento-Tormo et al. 2018)

Discussion

Given what we learned today, what could go wrong with this approach?

Hint 1: Clustering uses the same expression matrix as . . .

Hint 2: But L-R genes may not overlap with the genes that . . .

Discussion

What will be the underlying generative model?

Can we fix this? Could count splitting help?

- Split counts: $X = X^{\text{train}} + X^{\text{test}}$
- Cluster on X^{train}, test L-R pairs on X^{test}

But wait ...

- L-R scores need **mean expression** — do we have enough signal in X^{test}?
- Ligand in cluster A, receptor in cluster B — we need counts from **different cells**, not the same cell split twice
- What if the clustering itself changes with only half the counts?

Takeaways

Graph-based clustering

- Modularity optimization (Louvain/Leiden) is the workhorse of single-cell clustering
- The resolution parameter controls granularity — there is no single “correct” value
- SBMs offer a principled alternative with automatic model selection

When clustering goes wrong

- Clustering then testing on the same data = **double dipping** $\rightarrow$ inflated p-values
- **sc-SHC**: build significance testing into the clustering itself
- **Count splitting**: exploit Poisson thinning to create independent train/test sets
- **RECALL**: use artificial null genes to calibrate and detect over-clustering

Today's lecture

No perfect clustering algorithm

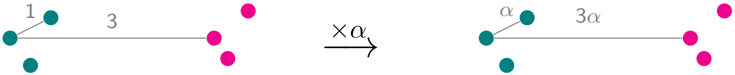
Impossibility Theorem (Kleinberg 2002): no clustering function f can simultaneously satisfy all three axioms:

1. **Scale invariance**: scaling all distances by $\alpha > 0$ doesn't change the clustering
2. **Richness**: every possible partition can be produced by some distance function
3. **Consistency**: shrinking within-cluster distances and expanding between-cluster distances doesn't change the clustering

Any algorithm must violate at least one:

Method	Scale inv.	Richness	Consistency
k-cluster (fixed k)	✓	×	✓
Distance threshold (fixed r)	×	✓	✓
Proportional ($\alpha \cdot d_{\max}$)	✓	✓	×

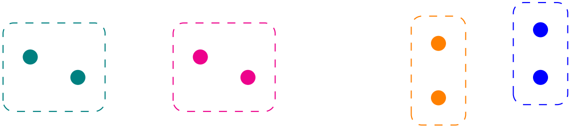
Axiom 1: Scale invariance



$f(d) = \{A, B\}$ $\qquad$ $f(\alpha \cdot d) = \{A, B\}$

Multiplying all distances by $\alpha > 0$ should not change the partition.

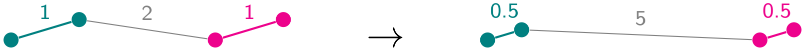
Axiom 2: Richness



$d_1 \to \{AB, CD\}$ $\qquad$ $d_2 \to \{AC, BD\}$

For **any** partition of n points, there exists some distance function d that produces it.

Axiom 3: Consistency



$f(d) = \{A, B\}$ $\qquad$ $f(d') = \{A, B\}$?

If we make clusters “tighter” and “farther apart,” the same partition should remain.

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